

Institute of Actuaries of India

Subject CT4 – Models

October 2014 Examination

INDICATIVE SOLUTION

Solution 1:-

- (i) (a) Let $Y_1, Y_2, \dots, Y_j, \dots$ be a sequence of independent and identically distributed random variables with

$$P(Y_j = 1) = P(Y_j = -1) = \frac{1}{2}$$

and define

$$X_n = \sum_{j=1}^n Y_j$$

Then $\{X_n\}_{n=1}^{\infty}$ constitutes a symmetric simple random walk.

- (b) Let N_t be a Poisson process, $t \geq 0$ and let $Y_1, Y_2, \dots, Y_j, \dots$ be a sequence of i.i.d. random variables. Then a compound Poisson process is defined by

$$X_t = \sum_{j=1}^{N_t} Y_j, \quad t \geq 0$$

[2]

- (ii) (a) A simple random walk operates on discrete time and has a discrete state space (the set of all integers, $\mathbb{Z}$).

- (b) A compound Poisson process operates on continuous time. It has a discrete or continuous state space depending on whether the variables Y_j are discrete or continuous respectively

[2]

- (iii) (a) For a process to be strictly stationary, the joint distribution of $X_{t_1}, X_{t_2}, \dots, X_{t_n}$ and $X_{t+t_1}, X_{t+t_2}, \dots, X_{t+t_n}$ are identical for all $t, t_1, t_2, \dots, t_n$ in $\mathbb{J}$ and all integers n .

This means that the statistical properties of the process remain unchanged over time.

- (b) Because strict stationarity is difficult to test fully in real life, we also use the less stringent condition of weak stationarity.

Weak stationarity requires that the mean of the process, $E[X_t] = m(t)$, is constant

and

the covariance, $E[(X_s - m(s))(X_t - m(t))]$, depends only on the time difference $t - s$.

[3]

[7 Marks]

Solution 2:-

(i) Consider a small time interval dt .

The probability of an arrival from the first process in time dt is $\lambda \cdot dt + o(dt)$ and the probability of a arrival from the second process in time dt is $\mu \cdot dt + o(dt)$.

The arrival probability for the sum of the processes in dt is therefore

$$(\lambda + \mu) \cdot dt + o(dt)$$

This is by definition a Poisson process with rate $(\lambda + \mu)$.

[2]

The Moment Generating Function or the Probability Generating Function of a Poisson distribution solution would also be acceptable.

(ii) (a) A jump chain is formed by recording the state of a Markov jump process only at the instant when a transition has just been made.

The jump chain is in itself a Markov chain.

(b) The outcome of the jump chain can only differ from that of the standard Markov chain if the jump process enters an absorbing state.

As the jump process will make no further transitions once it enters an absorbing state, the jump chain “stops”.

It is possible to model the jump chain as though transitions continue to occur but the chain continues to occupy the same state.

[4]

(iii) The possible states are 0 to N taxi booking counters in use with no passengers queuing, and

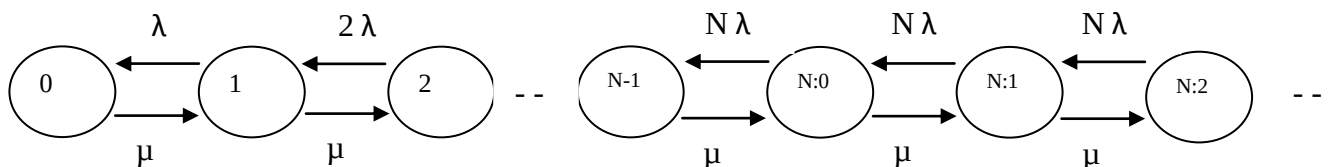
N counters in use with 0, 1, 2, passengers in the queue.

When all counters are occupied and there are M passengers in the queue denote the state as N:M.

State space is:

$\{0, 1, 2, \dots, N-1, N:0, N:1, N:2, \dots\}$

Transition diagram:



[3]

(iv) Kolmogorov forward equations in component form are:

$$\begin{aligned}\frac{d}{dt} P_0(t) &= \lambda P_1(t) - \mu P_0(t) \\ \frac{d}{dt} P_r(t) &= \lambda(r+1)P_{r+1}(t) + \mu P_{r-1}(t) - (\lambda r + \mu)P_r(t) \quad r+1 \leq N \\ \frac{d}{dt} P_{N:0}(t) &= \lambda N P_{N:1}(t) + \mu P_{N-1}(t) - (\lambda N + \mu)P_{N:0}(t) \\ \frac{d}{dt} P_{N:m}(t) &= \lambda N P_{N:m+1}(t) + \mu P_{N:m-1}(t) - (\lambda N + \mu)P_{N:m}(t) \quad m \geq 1\end{aligned}$$

[2]

(v) Poisson process is usually suitable for arrivals at a service point.

Rate may be time inhomogeneous because some passengers may have to wait for baggage arrival before able to book the taxi.

So a time-inhomogeneous Poisson process may be better.

[2]

Passengers may book taxis in family groups rather than individually and there is likely to be a minimum time for booking a taxi due to standard travel questions and billing details, so exponential distribution may not hold.

(vi) (a) The transition matrix is:

$$\begin{bmatrix} 0 & 1 & & & & & \\ \frac{\lambda}{\lambda + \mu} & 0 & \frac{\lambda}{\lambda + \mu} & & & & \\ & \frac{2\lambda}{2\lambda + \mu} & 0 & \frac{\lambda}{2\lambda + \mu} & & & \\ & & \ddots & \ddots & \ddots & & \\ & & & \frac{N\lambda}{N\lambda + \mu} & 0 & \frac{\lambda}{N\lambda + \mu} & \\ & & & & \frac{N\lambda}{N\lambda + \mu} & 0 & \frac{\lambda}{N\lambda + \mu} \\ & & & & & \ddots & \ddots \end{bmatrix}$$

(b) This is the probability that all the first N transitions are to the right in the transition diagram.

The probability of each transition is given by the elements in the upper half of the jump chain transition matrix in (vi)(a).

Required probability is therefore

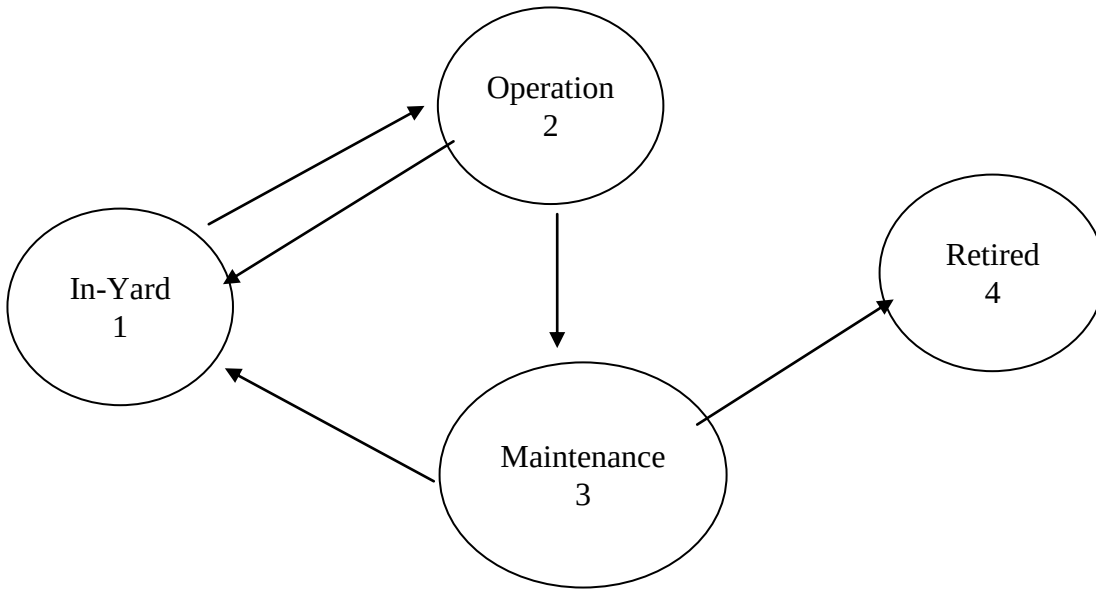
$$\mu^{N-1} \prod_{i=1}^{N-1} \frac{1}{i\lambda + \mu}$$

[4]

[17 Marks]

Solution 3:-

(i) The Multiple-state model diagram is:



[1]

(ii) Using the numbering of the states above, let the probability that an Engine who is in state i at time x will be in state j at time $x + t$ be ${}_t p_x^{ij}$.

Using the Markov property, and conditioning on the state occupied at time $x + t$, we can write:

$${}_{t+dt} p_x^{12} = {}_t p_x^{11} {}_{dt} p_{x+t}^{12} + {}_t p_x^{12} {}_{dt} p_{x+t}^{22}$$

Using the law of total probability, ${}_{dt} p_{x+t}^{22} = 1 - {}_{dt} p_{x+t}^{21} - {}_{dt} p_{x+t}^{23}$, so that

$${}_{t+dt} p_x^{12} = {}_t p_x^{11} {}_{dt} p_{x+t}^{12} + {}_t p_x^{12} (1 - {}_{dt} p_{x+t}^{21} - {}_{dt} p_{x+t}^{23})$$

Let the transition rate from state i to state j at time $x+t$ be μ_{x+t}^{ij}

Assume that ${}_{dt} p_{x+t}^{ij} = \mu_{x+t}^{ij} dt + o(dt)$, $i \neq j$

where

$$\lim_{dt \rightarrow 0^+} \frac{o(dt)}{dt} = 0$$

Substituting for the ${}_t p_x^{ij}$ in the equation above produces

$${}_{t+dt} p_x^{12} = {}_t p_x^{11} \mu_{x+t}^{12} dt + {}_t p_x^{12} (1 - \mu_{x+t}^{21} dt - \mu_{x+t}^{23} dt) + o(dt)$$

Therefore

$${}_{t+dt} p_x^{12} - {}_t p_x^{12} = {}_t p_x^{11} \mu_{x+t}^{12} dt - {}_t p_x^{12} \mu_{x+t}^{21} dt - {}_t p_x^{12} \mu_{x+t}^{23} dt + o(dt)$$

$$\lim_{dt \rightarrow 0^+} \frac{{}_{t+dt} p_x^{12} - {}_t p_x^{12}}{dt} = {}_t p_x^{11} \mu_{x+t}^{12} - {}_t p_x^{12} \mu_{x+t}^{21} - {}_t p_x^{12} \mu_{x+t}^{23}$$

So

$$\frac{d}{dt} {}_t p_x^{12} = {}_t p_x^{11} \mu_{x+t}^{12} - {}_t p_x^{12} \mu_{x+t}^{21} - {}_t p_x^{12} \mu_{x+t}^{23}$$

[5]

(iii) Let the waiting time in state i be v_i , and the number of transitions from state i to state j be d_{ij} and the transition intensity from state i to state j is μ_{ij}

Then the likelihood, L , may be written

$$L = K \exp [-v_1(\mu_{12}) - v_2(\mu_{21} + \mu_{23}) - v_3(\mu_{31} + \mu_{34})] \mu_{12}^{d_{12}} \mu_{23}^{d_{23}} \mu_{21}^{d_{21}} \mu_{31}^{d_{31}} \mu_{34}^{d_{34}}$$

[3]

(iv) The Logarithm of the likelihood is

$$\log_e L = \log_e K - v_1 \mu_{12} - v_2(\mu_{21} + \mu_{23}) - v_3(\mu_{31} + \mu_{34}) + d_{12} \log_e \mu_{12} + d_{23} \log_e \mu_{23} + d_{21} \log_e \mu_{21} + d_{31} \log_e \mu_{31} + d_{34} \log_e \mu_{34}$$

Differentiating with respect to μ_{12} ,

$$\frac{\partial L}{\partial \mu_{12}} = -v_1 + \frac{d_{12}}{\mu_{12}}$$

Setting this equal to 0 and solving for μ_{12} gives

$$\hat{\mu}_{12} = \frac{d_{12}}{v_1}$$

This is a maximum because $\frac{\partial^2 L}{\partial \mu_{12}^2} = -\frac{d_{12}}{\mu_{12}^2}$ which is negative.

[2]
[11 Marks]

Solution 4:-

It has come to our observation that determining the exact solution was time consuming and hence examiners were requested to be lenient while marking this question.

(i)

Let P be the transition matrix. We have:

$$P = \begin{bmatrix} x & 1-x & 0 & 0 \\ x & 0 & 1-x & 0 \\ 0.25 & x-0.25 & 0 & 1-x \\ 0.25 & 0 & x-0.25 & 1-x \end{bmatrix}$$

Let π be the long term stationary distribution.

Let a , b and c be the long-term probability of having 0%, 20% and 40% discount respectively. Since the long-term probability of having a 50% discount is the same as 0% discount, this would also be a .

Using, $\pi = \pi P$ we get:

$$\begin{bmatrix} a & b & c & a \end{bmatrix} = \begin{bmatrix} a & b & c & a \end{bmatrix} P$$

We have $2a + b + c = 1$

Also,

$$a = (a+b) x + \frac{(a+c)}{4}$$

$$\text{and } a = (a+c)(1-x)$$

Solving the equations we get:

$$\Rightarrow x = \frac{3}{4}(a + c)$$

Further,

$$b = a(1-x) + c(x - 0.25)$$

$$\text{and } c = b(1-x) + a(x - 0.25)$$

Solving the above equations we get $a = 0.3066$, $b = 0.2153$ and $c = 0.1714$ and $x = 0.1085$

The probability of making exactly one claim, therefore, is 10.85%.

[4]

(ii) The transition matrix is:

$$M = \begin{vmatrix} 0.3585 & 0.6415 & - & - \\ 0.3585 & - & 0.6415 & - \\ 0.2500 & 0.1085 & - & 0.6415 \\ 0.2500 & - & 0.1085 & 0.6415 \end{vmatrix}$$

$$M^2 = \begin{vmatrix} 0.3585 & 0.2299 & 0.4114 & 0 \\ 0.2889 & 0.2996 & 0 & 0.4114 \\ 0.2889 & 0.1603 & 0.1392 & 0.4114 \\ 0.2771 & 0.1721 & 0.0696 & 0.4811 \end{vmatrix}$$

The probability that a policyholder at the 20% level this year will be at the 50% level after three years is : $(0.3585*0)+(0*0.4114)+(0.6415*0.4114)+(0*0.4811) = 0.2640$.

[2]

(iii) The probability over a 20-year period will be based on the stationary distribution. The stationary probability of being in "40% Discount" state is 17.14%. Thus, the probability that a person with 0% discount would be enjoying a 40% discount 20 years from now would be 17.14%.

[3]

(iv)

(All figures in INR)

	No Claims Made	Both Claims recovered	1 Claim Recovered (Rs 7,500)	1 Claim Recovered (Rs 10,000)
1 - Year Gross Premium	20,000	20,000	20,000	20,000
Total Premium - 3 years (1)	60,000	60,000	60,000	60,000
NCD %	50%	0%	20%	20%
Discount (2)	30,000	-	12,000	12,000
Claim recovery from Insurer (3)	-	17,500	7,500	10,000
Out-Of-Pocket Claim Amount (4)	17,500	-	10,000	7,500
Net Cash flow = - (1 - 2 + 3 - 4)	(47,500)	(42,500)	(50,500)	(45,500)

From the table above, it can be seen that the net cash flow is minimum (absolute value) when both the claims are recovered by Mr. John. Thus no savings are achieved in terms of protecting the NCD by not claiming either or both the claims.

[5]
[14 Marks]

Solution 5:-

i)

$$U_x = 1/(w-x) \quad \text{where } 0 < x < w;$$

$$\begin{aligned} S_x &= \exp \left[- \int U_y dy \right] \\ &= \exp \left[- \int (1/(w-x)) dy \right] \\ &= \exp \left[-\log (1/(w-x)) \right] \\ &= (w-x) / w \end{aligned}$$

$$\begin{aligned} E[T(x)] &= \int_0^t t p_x U_{x+t} dt & 0 < t < w-x; \\ &= \int_0^t S_{x+t}/S_x U_{x+t} dt & 0 < t < w-x; \\ &= \int_0^t 1/(w-x) dt & 0 < t < w-x; \\ &= (w-x)/2 \end{aligned}$$

$$V[T(x)] = E[T(x)^2] - E[T(x)]^2$$

$$E[T(x)^2] = \int_0^{w-x} t^2 t p_x U_{x+t} dt \quad 0 < t < w-x ;$$

$$= (w-x)^2/3$$

$$V[T(x)] = (w-x)^2/3 - (w-x)/2$$

$$= (w-x)^2/12$$

$$V[T(15)] = (w-15)^2/12$$

$$= 533.4$$

$$\text{Therefore } w = 95$$

Complete expectation of life at age 25

$$(w-25)/2 = 35$$

[4]

ii)

The function assumes that people die in a linear fashion which is not exactly true due to infant mortality or higher probability of accidents at early ages leading to humped hazard curve.

[1]

[5 Marks]

Solution 6:-

i) The Cox regression model is an example of a semi-parametric approach, in which we do not pre-constrain the precise form of the hazard function.

- Fully parametric models are difficult to apply without foreknowledge of the form of the hazard function, which might be the very object of the study. For this reason a semi-parametric approach can be more popular
- Non-parametric approaches are limited in their ability to deal with some important questions in survival analysis, such as the effect of covariates on survival.

[2]

ii)

J	Ordered failure time X_i	Number at risk		Likelihood Contribution
		Group 0	Group 1	
1	3	5	5	$e^B/[5 + 5e^B]$
2	5	4	4	$e^B/[4 + 4e^B]$
3	6	4	2	$e^B/[6 + 8e^B + e^{2B}]^*$
4	9	2	0	$e^0/2 = 1/2$

* The tie occurs at $t = 6$, when $R(t_j) = \{Z = 0 : (6, 8+, 9, 10+), Z = 1 : (6, 8+)\}$

Of the $(6 | 2) = 15$ possible pairs of subjects at risk at $t=6$, there are 6 pairs formed where both are from group 0 ($S_j = 0$), 8 pairs formed with one in each group ($S_j = 1$), and 1 pairs formed with both in group 1 ($S_j = 2$).

The partial likelihood is given by

$$e^B / [5 + 5e^B] * e^B / [4 + 4e^B] * e^B / [6 + 8e^B + e^{2B}] * 1/2$$

Taking logs to find the partial log-likelihood:

$$\text{Log } L(B) =$$

$$3B - \log [5 + 5e^B] - \log [4 + 4e^B] - \log [6 + 8e^B + e^{2B}] + \log (1/2)$$

Alternative Solution using Breslow approximation:

The partial likelihood is given by

$$e^B / [5 + 5e^B] * e^B / [4 + 4e^B] * e^B / [(4 + 2e^B)^2] * 1/2$$

$$\text{Log } L(B) = 3B - \log [5 + 5e^B] - \log [4 + 4e^B] - 2 \log [4 + 2e^B] + \log (1/2)$$

[5]

iii)

We need to solve the equation $\text{Log } L'(B) = 0$

$$3B - \log [5 + 5e^B] - \log [4 + 4e^B] - \log [6 + 8e^B + e^{2B}] + \log (1/2)$$

Differentiating

$$3 - [e^B / (1 + e^B)] - [e^B / (1 + e^B)] - [(8e^B + 2e^{2B}) / (6 + 8e^B + e^{2B})]$$

$$3 - 2 [e^B / (1 + e^B)] - [(8e^B + 2e^{2B}) / (6 + 8e^B + e^{2B})]$$

Dividing by 2 we get for ease of calculation:

$$1.5 - [e^B / (1 + e^B)] - [(4e^B + e^{2B}) / (6 + 8e^B + e^{2B})]$$

If we write $f(x) = \text{Log } L'(B)$, we can apply the Newton-Raphson method, starting with $k = 1.80$. To do this, we need the second derivative:

$$\text{Log } L''(B) =$$

$$- [e^B(1 + e^B) - e^{2B}] / [(1 + e^B)^2] - [(4e^B + 2e^{2B}) * (6 + 8e^B + e^{2B}) - (4e^B + e^{2B}) * (8e^B + 2e^{2B})] / (6 + 8e^B + e^{2B})^2]$$

$$B^* = x - [f(x)/f'(x)] = 1.8 - [-0.03/-0.18] = 1.65$$

Alternative Solution using Breslow approximation:

$$\begin{aligned} & \text{Differentiating } 3B - \log [5 + 5e^B] - \log [4 + 4e^B] - 2 \log [4 + 2e^B] + \log (1/2) \quad 3 - [e^B/(1 + e^B)] - [e^B/(1 + e^B)] - [2e^B/(2 + e^B)] \\ & 3 - 2 * [e^B/(1 + e^B)] - [2e^B/(2 + e^B)] \end{aligned}$$

Dividing by 2 we get for ease of calculation: 1.5 –

$$[e^B/(1 + e^B)] - [e^B/(2 + e^B)]$$

If we write $f(x) = \log L'(B)$, we can apply the Newton-Raphson method, starting with $k = 1.80$. To do this, we need the second derivative:

$$\log L''(B) =$$

$$- [e^B(1 + e^B) - e^{2B}] / [(1 + e^B)^2] - [e^B(2 + e^B) - e^{2B}] / [(2 + e^B)^2]$$

$$B^* = x - [f(x)/f'(x)] = 1.8 - [-0.11/-0.31] = 1.44$$

[6]

iv)

$$\text{Taking MLE } B = 1.65$$

$$\log L''(1.65) = -0.18$$

MLEs are asymptotically unbiased and normally distributed with variance equal to the Cramér-Rao lower bound.

$$\text{MLE } B \sim N(1.65, 1/0.18)$$

$$\text{MLE } B \sim N(1.65, 5.55)$$

The test we require here is: $H_0 : B = 1$

versus $H_1 : B \neq 1$ The test statistic is:

$$0.65 / (5.55)^{0.5} = 0.27$$

This is not significant at the 5% level. So there is no reason to think that the rate differs by group.

Alternative Solution using Breslow approximation:

$$\text{Taking MLE } B = 1.44$$

$$\log L''(1.44) = -0.37$$

$$\text{MLE } B \sim N(1.44, 1/0.37)$$

$$\text{MLE } B \sim N(1.44, 2.7)$$

The test statistic is:

$$0.44 / (2.7)^{0.5} = 0.26$$

This is not significant at the 5% level. So there is no reason to think that the rate differs by group

[2]
[15 Marks]

Solution 7:-

i)

- The life assurance companies may have data in this form and hence it might be far easier than working with lives assured
- This would result in an unbiased estimate of the rate, provided there is no correlation between deaths and the number of policies covering a life.
- The presence of duplicates will increase the variance of the number of claims; we could make allowance for the increased variances in tests of a graduation.

[3]

(ii) Let D_i be the number of deaths among the $\pi_i N$ lives each with i policies, and let C_i be the number of claims among the same lives (i.e. $C_i = iD_i$).

We can say that:

$D_i \sim \text{Binomial}(\pi_i N, q_x)$; because we have independence of deaths

Therefore:

$$\begin{aligned} \text{Var}[C] &= \text{Var}[\sum C_i] \\ &= \text{Var}[\sum i D_i] \\ &= \sum i^2 \text{Var}[D_i] \quad (\text{independence of deaths}) \\ &= \sum i^2 \pi_i N q_x (1 - q_x) \end{aligned}$$

If we observed $\sum i \pi_i N$ independent lives (or policies) we would have:

$$\begin{aligned} E[C] &= \sum i \pi_i N q_x \\ \text{Var}[C] &= \sum i \pi_i N q_x (1 - q_x) \end{aligned}$$

So, the effect of duplicate policies is to increase the variance of the number of claims, in the ratio

$$r = \frac{\sum i^2 \pi_i}{\sum i \pi_i}$$

The proportions π_i ($i = 1, 2, 3, \dots$) depend on age, so we get a different variance ratio r at each age.

[4]

[7 Marks]

Solution 8:-

i)

Let t_i be the observed duration for part i and let d_i be equal to 1 if part i failed, and 0 otherwise. For those parts that did not fail, $t_i = t_s$.

Contribution to the likelihood for parts that fail is $f(t_i) = \mu \exp(-\mu t_i)$, and contribution for those parts that do not fail before time t_s is $S(t_i) = \exp(-\mu t_i)$. Putting these together gives

$$L = \prod_i [\mu \exp(-\mu t_i)]^{d_i} [\exp(-\mu t_i)]^{1-d_i} = \prod_i \mu^{d_i} \exp(-\mu t_i)$$

$$\log_e L = \sum_i d_i \log_e \mu - \mu \sum_i t_i = N_f \log_e \mu - \mu \sum_i t_i$$

We can now find the maximum-likelihood estimator $\hat{\mu}$ by differentiating with respect to μ and setting the derivative equal to zero

We get

$$\hat{\mu} = N_f / \sum_{i=1}^n t_i$$

[5]

ii)

$$N = 50$$

$$N_f = 20$$

$$\sum t_i = 20 * 1.2 + 30 * 2 = 84 \text{ weeks}$$

$$\begin{aligned} \hat{\mu} &= N_f / \sum t_i \\ &= 20/84 \\ &= 0.238/\text{week} \end{aligned}$$

Hence the mean time till failure is estimated to be $1/\hat{\mu} = 1/0.238$

$$= 4.2 \text{ weeks.}$$

[2]
[7 Marks]

Solution 9:-

i)

a) Maximum smoothness would be achieved by ignoring the plotted estimated rates and drawing a graduation curve which is very smooth, e.g. straight line. The deviations between the rates read from such a curve and the observed rates are likely to be very large. The graduation curve will be very smooth but have poor adherence to data.

On the other hand joining up a plot of the estimated values will give perfect adherence to data but is likely to produce a curve with rapidly changing curvature which would not satisfy the

smoothness criteria.

b) Graduation aims to resolve these conflicts by striking a fair balance between the adherence to data and smooth progression of rates.

Graduated rates can be obtained by many methods, some ensure smoothness, e.g. graduation by a mathematical form (the chosen functional form will ensure smoothness), reference to a standard table (a simple relationship with an already smooth set of standard table rates will ensure smoothness). In this case the graduated rates just need to satisfy tests of adherence to data.

Graphical graduation does not ensure smoothness, so graduated rates must be checked for smoothness and adherence to data. The graduation process must be repeated until both criteria are satisfied.

[4]

ii) a) Provided a formula with a small number of parameters is chosen the resulting graduation will be acceptably smooth.

(b) The graduation should be tested for smoothness using the third differences of the graduated rates which should be small in magnitude and progress regularly.

A further iterative process, which involves manual adjustment of the graduation (called 'hand-polishing') is sometimes necessary to ensure smoothness.

[2]

iii) By rearranging we get

$$\log_e(qx/px) = a+b*x$$

$$(qx/px) = \exp (a+b*x)$$

$$qx/(1-qx) = \exp (a+b*x)$$

$$qx = 1/(1+\exp(-(a+bx)))$$

Age	a+bx	qx	E	A	A-E	(A-E) ² /E
30	-7.0140	0.000898	224.60	245.00	20.40	1.85
31	-6.8978	0.001009	252.25	276.00	23.75	2.24
32	-6.7816	0.001133	283.29	313.00	29.71	3.12
33	-6.6654	0.001273	318.16	323.00	4.84	0.07
34	-6.5492	0.001429	357.30	339.00	-18.30	0.94
Total			1,435.60	1,496.00	60.40	8.22

Chi-square value: 8.22

However we have used two estimated parameters. So the number of degrees of freedom to use is 5-2 = 3. The upper 5% point of chi square (3) distribution is 7.815. So on the basis of chi square test, our hypothesis of good fit is rejected.

[5]
[11 Marks]

Solution 10:-

Please award 1/2 marks for each reasonable point. The requirements of good model are:

- The model should be able to produce past observed trends in mortality
- The model should be capable of producing sample paths of future rates of mortality i.e. stochastic in nature
- Is capable of projecting the mortality rates for all ages and for male and female separately
- Capable of producing cohort effect to reflect that people born in different time period might show different rates of mortality improvement
- May reflect the fact that the mortality improvements in the future might not follow the historic observed patterns
- Is simple to use and the output easy to communicate
- Should be capable of producing smooth set of mortality rates
- Calibration of the parameters should not be very complex and time consuming

Data related challenges:

- Not much data is available for higher ages i.e. ages 85 and above
- Any inaccuracies or inconsistencies in the data could have a significant effect the projections
- The underlying data will be heterogeneous in nature belonging to various geographical locations and socio-economic classes which might exhibit different mortality behaviour.
- The underlying cause of death might not be correctly reflected in the data

[6]
